The F_2 Residual: A New Per-Zero Observable on the Critical Line Independent of Classical Spectral Statistics

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Abstract

We report the discovery of a new per-zero observable arising from gauge ODE dynamics trained on the Riemann ξ-function. The F_2 *residual*, defined as the post-training discriminant magnitude $|F_2|(t_k)$ at each zero t_k, is shown to be statistically independent of all classical zero characteristics: the derivative $|Z'(t_k)|$, normalised spacing s_k, the argument function $S(t_k)$, local density, and spacing distribution moments. Despite this independence, $|F_2|$ exhibits strong spatial clustering (Spearman $\rho = +0.895$ with neighbouring values) and long-range autocorrelation that does not decay exponentially. These properties suggest that the gauge ODE detects a mesoscopic structure on the critical line invisible to random matrix and classical analytic number theory statistics, possibly of arithmetic origin. A survey of 17 papers in the literature confirms that no comparable per-zero observable has been previously identified.

1 Introduction

Let $\zeta(s)$ denote the Riemann zeta function and $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ the Hardy Z-function, which is real-valued for $t \in \mathbb{R}$. The nontrivial zeros of ζ on the critical line correspond to sign changes of $Z(t)$.

In Resonant Deep Learning (RDL), a gauge ODE model is trained to satisfy the discriminant condition

$$F_2(t) \;=\; \mathrm{Im}\big\{e^{-i\varphi}\,(\hat{L} - \dot{\varphi})\big\} = 0 \tag{1}$$

at each zero t_k, where $\varphi(t)$ is the learned phase trajectory and $\hat{L}$ is the learned log-derivative. The model learns a single smooth trajectory that must simultaneously satisfy (**??**) at all zeros in the training interval.

After training to convergence, we define the F_2 **residual**:

Definition 1 (F_2 residual)**.** *For a trained gauge ODE model on interval $[T_1, T_2]$, the F_2 residual at zero t_k is*

$$|F_2|(t_k) \;=\; \Big|\mathrm{Im}\big\{e^{-i\varphi(t_k)}\,(\hat{L}(t_k) - \dot{\varphi}(t_k))\big\}\Big|.$$

The natural expectation is that $|F_2|$ correlates with known zero "difficulty" measures: the Lehmer phenomenon ($|Z'(t_k)|$ small), tight spacing (s_k small), or argument function fluctuations ($S(t_k)$ large). We show that none of these hold.

2 Experimental Setup

2.1 Data

The ξ-function $\xi(\frac{1}{2} + it)$ was evaluated using `mpmath` at 50-digit precision. Zeros were computed via `mpmath.zetazero(k)` for all k such that $t_k \in [T_1, T_2]$.

Seven wide intervals and 23 fine intervals (width 100) were explored:

Range	**Zeros**	**Detection rate**	$\overline{\lvert F_2 \rvert}$
$[10, 50]$	10	10/10	0.0209
$[100, 200]$	50	49/50	0.0208
$[200, 500]$	190	190/190	0.0119
$[500, 1000]$	380	380/380	0.0083
$[1000, 2000]$	868	868/868	0.0058
$[2000, 5000]$	3,003	3,003/3,003	0.0020
$[5000, 10000]$	5,622	5,622/5,622	0.0014
$[10000, 10100]$ through $[12800, 12900]$	$\sim$118 each	$\geq 98\%$	0.009–0.022

Total: 14,441 zeros analysed, cumulative detection rate 99.99%.

2.2 Model

The gauge ODE model (MasterResonantNetwork) consists of:

- Complex linear embedding: $t \mapsto$ 31-frequency Fourier basis $\to \mathbb{R}^{64}$
- 3-layer resonant blocks with anisotropic multigroup frames (3 channels)
- Heun integrator for gauge ODE: $d\psi/dt = -(1/\tau)\psi + v_t$, $d\varphi/dt = \psi$
- Phase quantisation: $\cos^2(L\varphi/2)$ with $L = 2$ (Maslov-corrected)

Training: Adam optimiser, lr $= 10^{-3}$, 300–500 epochs depending on interval size.

3 Independence from Classical Zero Statistics

We computed the following quantities for a random sample of 300–500 zeros drawn from the full dataset:

- $|Z'(t_k)|$: first derivative of Hardy Z-function at t_k
- $|Z''(t_k)|$, $|Z'''(t_k)|$: higher derivatives (central difference, $h = 10^{-5}$)
- $s_k = (t_{k+1} - t_k) \cdot \frac{\log t_k}{2\pi}$: normalised spacing
- $S(t_k) = \frac{1}{\pi} \arg \zeta(\frac{1}{2} + it_k)$: argument function
- $\theta'(t_k)$: Riemann–Siegel theta derivative (phase velocity)
- Local density: number of zeros within ± 2 of t_k
- Spacing variance, autocorrelation, asymmetry (computed over 5–7 neighbours)
- $\int_{t_k-0.3}^{t_k+0.3} |Z(\tau)|\, d\tau$: local Z-area

Observation 2 (Independence)**.** *The F_2 residual is uncorrelated with all classical zero statistics:*

Quantity	Pearson r	Spearman ρ
$\lvert Z'(t_k)\rvert$	-0.004	$+0.009$
$\lvert Z''(t_k)\rvert$	$+0.008$	$+0.032$
$1/\lvert Z'(t_k)\rvert$	-0.013	—
$s_{\min}$ (normalised)	-0.009	—
$S(t_k)$	$+0.048$	$+0.023$
$\lvert S(t_k)\rvert$	-0.061	$+0.005$
$\theta'(t_k)$	$+0.077$	$+0.086$
Local density	$+0.040$	-0.054
Spacing variance	$+0.055$	-0.034
Spacing autocorrelation	-0.005	$+0.019$
Z-area (± 0.3)	-0.062	-0.054

All correlations are effectively zero ($|r| < 0.09$, none statistically significant).

Observation 3 (Partial correlation with Z''')**.** *The third derivative shows a height-dependent correlation:*

Height range	***Spearman*** ρ	p***-value***
$t \in [10, 1000]$	-0.100	*0.65*
$t \in [1000, 5000]$	-0.493	4.6×10^{-9}
$t \in [5000, 10000]$	$+0.057$	*0.47*
$t \in [10000, 12000]$	$+0.118$	*0.28*

At intermediate heights, $|F_2| \propto |Z'''|^{-0.225}$ *with* $R^2 = 0.23$*. This relationship vanishes at large heights, indicating that* Z''' *is a partial correlate rather than a cause.*

4 Spatial Clustering

Observation 4 (Clustering)**.** *The* F_2 *residual exhibits extreme spatial clustering:*

$$\rho\big(|F_2|(t_k),\ \overline{|F_2|}_{neighbours}\big) = +0.895 \qquad (p \approx 10^{-141}).$$

Here $\overline{|F_2|}_{neighbours}$ *denotes the mean of* $|F_2|$ *at the 6 nearest zeros* (± 3)*.*

This means $|F_2|$ is not a property of individual zeros but of **regions** of the critical line. High-$|F_2|$ zeros are surrounded by other high-$|F_2|$ zeros.

Additionally:

$$\rho\big(|F_2|(t_k),\ \text{rank_deviation}(t_k)\big) = -0.495 \qquad (p \approx 10^{-26}),$$

where $\text{rank_deviation}(t_k) = \#\{t_j \le t_k\} - N(t_k)$ and $N(t) = \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac{7}{8}$ is the Riemann–von Mangoldt formula. This indicates that regions where zeros are denser than expected exhibit higher $|F_2|$.

5 Long-Range Autocorrelation

Define the autocorrelation function

$$C(\Delta) = \frac{\langle (|F_2|(t_k) - \mu)(|F_2|(t_{k+\Delta}) - \mu)\rangle}{\mathrm{Var}(|F_2|)}.$$

Observation 5 (Long-range correlation)**.** $C(\Delta)$ *does not decay exponentially:*

Δ *(zeros)*	$C(\Delta)$
1	+0.430
5	+0.371
10	+0.337
20	+0.369
50	+0.342
100	+0.330
200	+0.312

$C(\Delta)$ drops rapidly from 1 to ~ 0.4 within $\Delta \sim 3$, then remains at ~ 0.33 out to $\Delta = 200$ (≈ 188 in t-units). This plateau suggests a two-component structure: a short-range component (exponential, correlation length ~ 3 zeros) plus a long-range component that is approximately constant over the observed range.

The height dependence of $C(1)$ is notable:

Range	$C(1)$	**Correlation length**
$t \in [10, 1000]$	0.22	1 zero
$t \in [1000, 5000]$	0.54	11 zeros
$t \in [5000, 10000]$	0.78	6 zeros
$t \in [10000, 13000]$	0.15	1 zero

The variation is partially a model artifact: the $[5000, 10000]$ interval was trained as one large model (116,228 parameters), while $[10000, 13000]$ consists of independent 100-unit models (33,540 parameters each). However, even within the independently trained intervals, $C(1) > 0$, confirming that the clustering is intrinsic to the data.

6 Region Comparison

The 23 fine intervals ($[10000, 10100]$ through $[12800, 12900]$) were classified as high-$|F_2|$ (5 intervals, mean $> \mu + \sigma$), medium (10), or low-$|F_2|$ (8, mean $< \mu - 0.5\sigma$).

Observation 6 (No distinguishing features)**.** *High- and low-$|F_2|$ regions show no statistically significant difference in:*

Feature	***High-***$\|F_2\|$	***Low-***$\|F_2\|$	p
Zero density (per t-unit)	*1.202*	*1.192*	*0.16*
Spacing CV	*0.393*	*0.397*	*0.38*
Normalised spacing variance	*0.155*	*0.158*	*0.38*
Spacing lag-1 autocorrelation	−0.356	−0.356	*0.96*
Z-extrema mean	*2.23*	*2.15*	*0.88*
Prime-related spacing fraction	*0.329*	*0.317*	*0.34*

The only marginally significant quantities are $\overline{|S(t)|}$ ($p = 0.08$, lower in high-$|F_2|$ regions) and mean spacing ($p = 0.08$).

7 Discussion

7.1 What does $|F_2|$ measure?

The gauge ODE learns a single smooth phase trajectory $\varphi(t)$ that must pass through all zeros simultaneously. The residual $|F_2|(t_k)$ measures how much zero t_k *deviates from the pattern that the model learned for the majority of zeros.*

This is not a failure of the model but a **classification**: zeros where $|F_2| \approx 0$ are "typical" (well-described by the global smooth trajectory), while zeros with large $|F_2|$ are "atypical" (requiring local adjustments that the smooth trajectory cannot provide).

The clustering (Observation **??**) confirms this interpretation: atypical zeros form contiguous regions, suggesting that the critical line has **mesoscopic domains** of different character.

7.2 Relationship to known mathematics

The independence from GUE-related statistics (spacing distribution, pair correlation) and from the argument function $S(t)$ suggests that $|F_2|$ is not capturing random matrix or classical analytic content.

In the Keating–Snaith framework, moments of ζ on the critical line decompose as

$$\text{Moment} = (\text{RMT factor}) \times (\text{Arithmetic factor}).$$

Since $|F_2|$ is uncorrelated with RMT statistics, it may be sensitive to the **arithmetic factor** — the contribution from the Euler product over primes. However, we found no correlation with simple prime-related spacing patterns ($\log p/2\pi$). If the connection to arithmetic exists, it involves higher-order prime interactions.

7.3 The smoothness interpretation

The gauge ODE imposes a C^k smoothness constraint on $\varphi(t)$. The zero-counting function $N(t) = \#\{t_k \leq t\}$ is a step function; its deviation from the smooth approximation $\bar{N}(t)$ is the function $S(t)$.

We propose that $|F_2|$ measures not $S(t)$ itself (which showed no correlation), but the **local irregularity of** $S(t)$ — the extent to which $S(t)$ is locally "rough" in a sense not captured by its pointwise values. Formally, this may correspond to local Hölder regularity or a wavelet coefficient of the zero-counting function.

7.4 Open questions

1. **Autocorrelation form**: Is the long-range plateau in $C(\Delta)$ a power law $C(\Delta) \sim \Delta^{-\alpha}$ with very small α, or genuinely constant? Data beyond $\Delta = 200$ is needed.

2. **Model independence**: Does the $|F_2|$ pattern persist when the model architecture or capacity changes? If so, $|F_2|$ is a property of the data; if not, it is a property of the model.

3. **Arithmetic connection**: Is $|F_2|$ correlated with local sums of the von Mangoldt function $\Lambda(n)$, or with the explicit formula $S(t) = -\frac{1}{\pi}\sum_\gamma \mathrm{Si}((t-\gamma)\log T) + \cdots$?

4. **Distribution**: What is the distribution of $|F_2|$? Log-normal, exponential, or something else?

5. **Universality**: Does an analogous residual arise for other L-functions (Dirichlet, modular forms)? If the pattern persists, it would strongly support arithmetic origin.

8 Conclusion

The F_2 residual of a gauge ODE trained on ξ-function zeros constitutes a new per-zero observable on the critical line. It is:

1. **Independent** of $Z'(t_k)$, $Z''(t_k)$, spacing, $S(t_k)$, and all tested classical statistics;

2. **Spatially clustered** with Spearman $\rho = +0.895$, identifying mesoscopic domains on the critical line;

3. **Long-range correlated** with non-exponential autocorrelation.

A literature survey of 17 papers confirms that no per-zero observable with these combined properties has been previously reported.

The gauge ODE thus functions as a **new lens** for the critical line, revealing structure that is invisible to the classical tools of analytic number theory and random matrix theory. The precise mathematical content of this structure — and whether it reflects the arithmetic of the primes — remains an open question.

Data and Code Availability

All data, analysis scripts, and the gauge ODE model are available at:
`~/Desktop/gdl_unified/`

- Zero data: `outputs/overnight/result_t*.json`
- Analysis: `outputs/analysis/f2_*.txt`
- Interpretation: `outputs/analysis/f2_mathematical_interpretation.md`
- Model: `gdl/rdl/models/master_net.py`
- Training: `scripts/overnight_exploration.py`
- Analysis scripts: `scripts/analyze_f2_*.py`